Ziteng Yang

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Education

Georgia Institute of Technology (Georgia Tech), Atlanta, GA, USA 2021.08 - 2027.03

Ph.D. candidate in Computer Science, advised by Dr. *Vivek Sarkar*

- **Thesis & Research Areas:** Formal Verification, Compilers
- **Graduate Courses:** Advanced Algorithms (convex optimization and sampling), High-Performance Computer Architecture (multi-core CPU simulator), Compiler Design (LLVM), Parallelizing Compiler, Software Analysis and Testing, **[Minor]** Advanced Graph Theory, Measure Theory (= Real Analysis)

Shanghai Jiao Tong University (SJTU), Shanghai, China 2017.09 – 2021.06

B.E. in Computer Science, with GPA=86; minor in Music

- **Thesis & Research Areas:** Math. Logic (first-order & modal), Program (Separation) Logic & Semantics
- **Selected Courses:** Theory of Computation (Turing, Gödel), Programming Language Theory (Typed λ), Machine Learning (NN, RL), Linux Kernel (memory pages, process/file system), Computer Network, Information Security, Abstract Algebra, Circuit Theory, Digital Electronics, Mechanics Theory, Game Design, Music Theory, Music Composition, Solfeggio, etc.

Industrial Experience

Amazon Web Services, Inc. (AWS), Automated Reasoning Group & Annapurna Labs

Formal verification of tensor compiler (***AWS Neuron***) code base assisted by theorem prover (***Lean***);

Applied Scientist Internship, Seattle, WA, USA Sep. ∼ Dec. 2026

Applied Scientist Internship, Seattle, WA, USA Jan. ∼ May. 2025

- Formal verification of internal compiler codebase assisted by Lean 4 prover and Z3 SMT solver

Meta Platforms, Inc., PyTorch & Triton Team

Compiler Engineer Internship, Menlo Park, CA, USA Jun. ∼ Aug. 2026

Compiler Engineer Internship, Bellevue, WA, USA May. ∼ Aug. 2025

Research & Engineering in tensor compiler (***Triton*** open-source language, based on LLVM/MLIR)

- Static & sound bitwise-equivalence checker of tensor IR and CUDA PTX / AMD GCN assembly
- Data-layout optimization for inner tree reduction
- Triton general matrix multiply (GEMM) GPU kernels bit-matching NVIDIA cuBLAS (closed source) with PyTorch Inductor integration and cuBLAS tensor dropping bug reproduced (AI agent workflow assisted)
- Debugging information preservation support in Triton's compiler

Research Papers

Z. Yang, J. Shirako, V. Sarkar, Fully Verified Instruction Scheduling, *OOPSLA'24*

Z. Yang, X. Yin and S. Li, Maximally permissive supervisor control of timed discrete-event systems under partial observation, in *21st IFAC World Congress, 2020*

Research Experience

Doctorate Thesis: Compiler Correctness for CPU and GPU Language Aug. 2021- Mar. 2027

Advisor: Dr. *Vivek Sarkar*, School of Computer Science, Georgia Tech.

- **[GPU Compiler Correctness] Taming bit-wise equivalence in GPU kernels and compilers**
- **[GPU Compiler Correctness] Translation Validation of Tile-level Tensor Language**
- **[CPU Compiler Correctness] Fully Verified Instruction Scheduling**
  [OOPSLA'24] Built the *first ever* foundational formal theory to fully verify instruction scheduling passes in a verified compiler and achieved the first fully verified scheduling pass in CompCert framework.

Bachelor Thesis on Program Logic, Proof Assistant and Formal Semantics 2020 – 2021

Advisor: Dr. *Qinxiang Cao*, John Hopcroft Center for Computer Science, SJTU.

- **Semantics Preservation for Annotation Verifier**: Designed and formalized (in Rocq) an extended semantics preservation framework for optimizing verified programs by exploiting information in Hoare assertions
- **Formalized Modal Logic**: Practice on Rocq by formally proving completeness theorem of Modal Logic.
- **Mathematical Logic Essentials**: Mathematical logic (First-order/Modal Logic) with soundness/completeness proof, and Hoare/(Concurrent) Separation Logic with soundness proof

Formal Control Theory of Timed Discrete-Event Systems Summer 2019

Advisor: Dr. *Xiang Yin*, Department of Automation, SJTU.

- [IFAC'20] Designed a synthesizing algorithm of safe and maximally permissive supervisor for Timed Discrete Event System and proved the language generated by the automata under the synthesized supervisor is safe yet maximal.

Academic Services

CGO 2027 Artifact Evaluation Committee
PLDI 2025 Artifact Evaluation Committee (Reviewed 3 artifacts)

Teaching Experience

Teaching Assistant, CS6390 Programming Languages, Georgia Tech, by *Vivek Sarkar* Spring 2023
Teaching Assistant, CS4510 Automata and Complexity, Georgia Tech, by *Joseph Jaeger* Fall 2022
Teaching Assistant, MA208 Discrete Mathematics (IEEE Honor Class), SJTU, by *Qinxiang Cao* Fall 2020
Teaching Assistant, MA239 Discrete Mathematics (Zhiyuan Honor Class), SJTU, by *Xiang Yin* Fall 2020

Skills

Programming Lanugages & Software Tools:

- **C/C++, Python, Java**: professionally skilled in industrial development and academic research
- **Lean 4, Rocq (previously Coq), OCaml, Z3 (SMT Solver)**: long-term research on formal verification
- **Tools / Libraries**: LLVM / MLIR (compiler IR tool-chain), Triton (tensor language & compiler), CUDA (Nvidia's GPU language), GDB& LLDB& CUDA-GDB (debugging tools), CompCert (a fully certified C compiler), Iris/VST (separating logic tools for verification), Linux Kernel Programming
- **Unity 5, Unreal Engine**: course projects & student org. experience on video game design
- **Logic Pro, Sibelius**: course projects experience on music composition

Non-Tech: Tennis, Piano

Honors and Scholarships

- Rongchang (荣昶) Scholarship for Science and Technology Innovation, Finalist, 10,000 CNY (20 finalists/winners among university) 2020
- Undergraduate Excellence Scholarship, 500 CNY 2018
- 1st Prize (50 winners per province) in National High School Mathematics League 2016